

Chapter 2

Section 2.1

Linear Partial Differential Equation of Order One

2.1.1 Quasi-linear partial differential equation of order one: If p and q are linear, then $\phi(x, y, z, p, q) = 0$ is called quasi-linear partial differential equation of order one. The general form of this equation is $P(x, y, z)p + Q(x, y, z)q = R(x, y, z)$.

Examples:

(i) $xzp + yzq = x^2y^2$,

(ii) $xyzp + (y - z)q = z^2 - xy$

2.1.2 Lagrange's method of solving the quasi-linear partial differential equation of order one namely
 $Pp + Qq = R$

Theorem 1. State and prove Lagrange's method of solving the linear partial differential equation of order one, namely $Pp + Qq = R$, where $p = \frac{\partial z}{\partial x}$, $q = \frac{\partial z}{\partial y}$ and P, Q, R are functions of x, y, z .

Statement. The general solution of the linear partial differential equation $Pp + Qq = R$ is $\phi(u, v) = 0$, where ϕ is an arbitrary function and $u(x, y, z) = c$ and $v(x, y, z) = c_1$ are solutions of equations $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$.

Proof : We treat z as dependent variable and x, y as independent variables.

So that $\frac{\partial z}{\partial x} = p, \frac{\partial z}{\partial y} = q, \frac{\partial y}{\partial x} = 0, \frac{\partial x}{\partial y} = 0$

Differentiating $\phi(u, v) = 0$ with respect to x , we get

$$\frac{\partial \phi}{\partial u} \left(\frac{\partial u}{\partial x} + p \frac{\partial u}{\partial z} \right) + \frac{\partial \phi}{\partial v} \left(\frac{\partial v}{\partial x} + p \frac{\partial v}{\partial z} \right) = 0 \quad \dots (1)$$

Again, differentiating $\phi(u, v) = 0$ with respect to y , we get

$$\frac{\partial \phi}{\partial u} \left(\frac{\partial u}{\partial y} + q \frac{\partial u}{\partial z} \right) + \frac{\partial \phi}{\partial v} \left(\frac{\partial v}{\partial y} + q \frac{\partial v}{\partial z} \right) = 0 \quad \dots (2)$$

Eliminating $\frac{\partial \phi}{\partial u}$ and $\frac{\partial \phi}{\partial v}$ from (1) and (2), we obtain

$$\begin{vmatrix} \frac{\partial u}{\partial x} + p \frac{\partial u}{\partial z} & \frac{\partial v}{\partial x} + p \frac{\partial v}{\partial z} \\ \frac{\partial u}{\partial y} + q \frac{\partial u}{\partial z} & \frac{\partial v}{\partial y} + q \frac{\partial v}{\partial z} \end{vmatrix} = 0$$
$$\Rightarrow \frac{\partial u}{\partial x} \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} + p \left(\frac{\partial u}{\partial z} \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial v}{\partial z} \right) + q \left(\frac{\partial u}{\partial x} \frac{\partial v}{\partial z} - \frac{\partial u}{\partial z} \frac{\partial v}{\partial x} \right) = 0$$

Putting $\lambda P = \frac{\partial u}{\partial z} \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial v}{\partial z}$, $\lambda Q = \frac{\partial u}{\partial x} \frac{\partial v}{\partial z} - \frac{\partial u}{\partial z} \frac{\partial v}{\partial x}$ and $\lambda R = \left(\frac{\partial u}{\partial x} \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} \right)$

$\therefore Pp + Qq = R$

Hence $\phi(u, v) = 0$ is the general solution of $Pp + Qq = R$

Again, consider $u(x, y, z) = c_1$ and $v(x, y, z) = c_2$

where c_1 and c_2 are arbitrary constants.

$\therefore du(x, y, z) = 0$ and $dv(x, y, z) = 0$

$$\Rightarrow \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy + \frac{\partial u}{\partial z} dz = 0 \text{ and } \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy + \frac{\partial v}{\partial z} dz = 0$$

Solving the above equations for dx, dy and dz , we get

$$\begin{aligned} \frac{dx}{\frac{\partial u}{\partial y} \frac{\partial v}{\partial z} - \frac{\partial u}{\partial z} \frac{\partial v}{\partial y}} &= \frac{dy}{\frac{\partial u}{\partial z} \frac{\partial v}{\partial x} - \frac{\partial u}{\partial x} \frac{\partial v}{\partial z}} = \frac{dz}{\frac{\partial u}{\partial x} \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial v}{\partial x}} \\ \Rightarrow \frac{dx}{P} &= \frac{dy}{Q} = \frac{dz}{R} \end{aligned} \quad \dots(3)$$

Thus $u(x, y, z) = c_1$ and $v(x, y, z) = c_2$ are solutions of (3). Hence, the theorem is proved.

[Note: The equations $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$ are called Lagrange's auxiliary (or subsidiary) equations for the partial differential equation $Pp + Qq = R$.]

2.1.3 Working rule for solving Lagrange's equation $Pp + Qq = R$

Step 1. Put the given linear partial differential equation of the first order in the standard form

$$Pp + Qq = R \quad \dots (1)$$

Step 2. Write down the Lagrange's auxiliary equations

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \quad \dots (2)$$

Step 3. Find two linear independent solutions of (2) with the help of any well known methods.

Step 4. If $u(x, y, z) = c_1$ and $v(x, y, z) = c_2$ are two independent solutions of $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$ then the general solution of $Pp + Qq = R$ is $\phi(u, v) = 0$ or $u = \phi(v)$, where ϕ is an arbitrary function.

Example-1 Find the general solution of the following partial differential equations:

(i) $y^2 zp - x^2 zq = x^2 y$

(ii) $yzp + xzq = xy$

(iii) $(y^2 z / x)p + xzq = y^2$

Solution

(i) The given equation is

$$y^2 zp - x^2 zq = x^2 y \quad \dots(1)$$

Here the Lagrange's auxiliary equations for (1) are

$$\frac{dx}{y^2z} = \frac{dy}{-x^2z} = \frac{dz}{x^2y} \quad \dots(2)$$

Taking first two fractions of (2), we get

$$\begin{aligned} \frac{dx}{y^2z} &= \frac{dy}{-x^2z} \\ \Rightarrow x^2dx + y^2dy &= 0 \\ \Rightarrow \frac{x^3}{3} + \frac{y^3}{3} &= \frac{c_1}{3} && \text{[Integrating both sides]} \\ \Rightarrow x^3 + y^3 &= c_1 && \dots(3) \end{aligned}$$

Again, taking last two fractions of (2), we get

$$\begin{aligned} \frac{dy}{-x^2z} &= \frac{dz}{x^2y} \\ \Rightarrow ydy + zdz &= 0 \\ \Rightarrow \frac{y^2}{2} + \frac{z^2}{2} &= \frac{c_2}{2} && \text{[Integrating both sides]} \\ \Rightarrow y^2 + z^2 &= c_2 && \dots(4) \end{aligned}$$

From (3) and (4), we obtain the required general solution $\phi(x^3 + y^3, y^2 + z^2) = 0$, where ϕ being an arbitrary function.

(ii) Given equation, $yzp + zxq = xy$... (1)

The Lagrange's auxiliary equation for (1) are

$$\frac{dx}{yz} = \frac{dy}{zx} = \frac{dz}{xy} \quad \dots (2)$$

Taking first two fractions of (2), we get

$$\begin{aligned} \frac{dx}{yz} &= \frac{dy}{zx} \\ \Rightarrow xdx - ydy &= 0 \\ \Rightarrow \frac{x^2}{2} - \frac{y^2}{2} &= \frac{c_1}{2} && \text{[Integrating both sides]} \\ \Rightarrow x^2 - y^2 &= c_1 && \dots(3) \end{aligned}$$

Again, taking last two fractions of (2), we get

$$\begin{aligned} \frac{dy}{zx} &= \frac{dz}{xy} \\ \Rightarrow ydy - zdz &= 0 \\ \Rightarrow \frac{y^2}{2} - \frac{z^2}{2} &= \frac{c_2}{2} && \text{[Integrating both sides]} \\ \Rightarrow y^2 - z^2 &= c_2 && \dots(4) \end{aligned}$$

From (3) and (4), we obtain the required general solution $\phi(x^2 - y^2, y^2 - z^2) = 0$, where ϕ being an arbitrary function.

$$(iii) \text{ Given equation, } (y^2 z/x)p + xzq = y^2 \quad \dots (1)$$

The Lagrange's auxiliary equations for (1) are

$$\frac{dx}{y^2 z/x} = \frac{dy}{xz} = \frac{dz}{y^2} \quad \dots (2)$$

Taking the first two fractions of (2), we get

$$\begin{aligned} \frac{dx}{y^2 z/x} &= \frac{dy}{xz} \\ \Rightarrow x^2 dx - y^2 dy &= 0 \\ \Rightarrow \frac{x^3}{3} - \frac{y^3}{3} &= \frac{c_1}{3} \quad [\text{Integrating both sides}] \\ \Rightarrow x^3 - y^3 &= c_1 \quad \dots(3) \end{aligned}$$

Again, taking the first and last fractions of (2), we get

$$\begin{aligned} \frac{dx}{y^2 z/x} &= \frac{dz}{y^2} \\ \Rightarrow x dx - z dz &= 0 \\ \Rightarrow \frac{x^2}{2} - \frac{z^2}{2} &= \frac{c_2}{2} \quad [\text{Integrating both sides}] \\ \Rightarrow x^2 - z^2 &= c_2 \quad \dots(4) \end{aligned}$$

From (3) and (4), we obtain the required general solution $\phi(x^3 - y^3, x^2 - z^2) = 0$, where ϕ being an arbitrary function.

Example-2 Solve the following partial differential equations:

$$(i) \quad xzp + yzq = xy \quad (ii) \quad p + 3q = 5z + \tan(y - 3x)$$

Solution

$$(i) \text{ Given equation } xzp + yzq = xy \quad \dots (1)$$

The Lagrange's auxiliary equations for (1) are

$$\frac{dx}{xz} = \frac{dy}{yz} = \frac{dz}{xy} \quad \dots (2)$$

Taking the first two fractions of (2), we get

$$\frac{dx}{x} = \frac{dy}{y}$$

Integrating, we obtain

$$\ln x - \ln y = \ln c_1$$

$$\Rightarrow \ln\left(\frac{x}{y}\right) = \ln c_1$$

$$\Rightarrow \frac{x}{y} = c_1 \quad \dots (3)$$

Again, taking the last two fractions of (2), we get

$$\frac{dy}{yz} = \frac{dz}{xy}$$

$$\Rightarrow \frac{dy}{z} = \frac{dz}{c_1 y} \quad [\because x = c_1 y]$$

$$\Rightarrow c_1 y dy - z dz = 0$$

Integrating, we obtain

$$c_1 \frac{y^2}{2} - \frac{z^2}{2} = \frac{c_2}{2}$$

$$\Rightarrow c_1 y^2 - z^2 = c_2$$

$$\Rightarrow xy - z^2 = c_2 \quad \dots (4)$$

From (3) and (4), we obtain the required general solution $\phi(x/y, xy - z^2) = 0$, where ϕ being an arbitrary function.

(ii) Given equation,

$$p + 3q = 5z + \tan(y - 3x) \quad \dots (1)$$

The Lagrange's auxiliary equations of (1) are

$$\frac{dx}{1} = \frac{dy}{3} = \frac{dz}{5z + \tan(y - 3x)} \quad \dots (2)$$

Taking the first two fractions of (2), we get

$$\frac{dx}{1} = \frac{dy}{3}$$

$$\Rightarrow 3dx = dy$$

Integrating, we obtain

$$y - 3x = c_1 \quad \dots (3)$$

Again, taking the firsts and third fractions of (2), we get

$$\frac{dx}{1} = \frac{dz}{5z + \tan(y - 3x)}$$

$$\Rightarrow dx = \frac{dz}{5z + \tan c_1} \quad [\because y - 3x = c_1]$$

Integrating, we obtain

$$x = \frac{1}{5} \ln(5z + \tan c_1) + c_2$$

$$\Rightarrow x - \frac{1}{5} \ln [5z + \tan(y - 3x)] = c_2 \quad \dots(4)$$

From (3) and (4), we obtain the required general solution $x - \frac{1}{5} \ln [5z + \tan(y - 3x)] = \phi(y - 3x)$, where ϕ being an arbitrary function.

Exercise-2.1

1. Find the general solution of the following partial differential equations:

(i) $x^2 p + y^2 q = z^2$

(ii) $a(p + q) = 1$

(iii) $p \tan x + q \tan y = \tan z$

(iv) $zxp + z(x - y)q = z^2$

(v) $(x - y)p + (x + y)q = z(y - x)$

(vi) $y^2 p - xyq = x(z - 2y)$

2. Solve the following partial differential equations:

(i) $xyp + y^2 q = zxy - 2x^2$

(ii) $p - q = 2x \sin(x + y)$

(iii) $xy^2 p - y^3 q + axz = 0$

(iv) $p - q = \frac{z}{x + y}$

(v) $(3x + y - z)p + (x + y - z)q = 2(z - y)$